Unified Source Theory (UST) v5: Theorems, Derivations, and GR-QFT Unified Matrix

Deterministic Operating Framework

1. Quantum Gravity (GR + QFT) Full Unified Equations

In the Unified Source Theory, macrocosmic gravity and microcosmic quantum states are deterministically locked together, weighted by the universal constitutional seal coefficient ($N_{s,q} = 0.63354460$). The root equations providing the full integration of GR and QFT are as follows:

1.1. Ontological GR-QM Equivalence and Lorentz Vacuum: Zero quantum entropy ($S_{Om} = 0$) is structurally equivalent to the stabilization of the spacetime geometry:

$$S_{Om} = 0 \iff G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} N_{s,q} T^{QM}_{\mu\nu} \tag{1}$$

1.2. Dynamic Quantum-Gravity Conservation (Modified Bianchi Identity - T40): Full energy-momentum conservation harboring quantum vacuum fluctuations:

$$\nabla_\mu \left(G^{\mu\nu} + \Lambda g^{\mu\nu} + \hbar Q^{\mu\nu} \right) = 0 \tag{2}$$

Here, $\hbar Q^{\mu\nu}$ is the 1-loop QFT correction acting on the system via the 4D Dirac operator and the Seeley-DeWitt ($a_2 = 1/17$) coefficient.

1.3. Ultimate Unified Master Equation (Omnium Field): The chaos-harmony regulation between Channel Q (Active Quantum Flow) and Channel C (Frozen Background):

$$\Omega_m = \underbrace{(\nabla\Phi \cdot \Lambda_{kin} \cdot Q_{res})}_{\text{Kinetic Sealing (Channel Q)}} + \underbrace{\ln(1 + \sigma\nabla\Phi)}_{\text{Harmonic Restoration (Channel C)}} \tag{3}$$

2. UST Theorem and Derivation Matrix (T1 - T43)

No	Analytical Equation / Formula	Constants Used and Derivation Origin	Addressed Field / Discipline
Root	$N_{s,q} = \frac{3-\sqrt{3}}{2} - \frac{\alpha}{17} \approx 0.6335446$	Seeley-DeWitt heat kernel QED correction derived from the 4D Dirac operator (DOF=17) via the geometric maximization of the Einstein tensor $G_{tt} = 1/4$.	Fundamental Physics / Entire Framework
T1	$\kappa \cdot dt = \pi \cdot N_{s,q} \approx 1.990339$	Matrix roots of the Decoherence-Free Subspace (DFS) operator.	Quantum Error Correction
T3	$S_{Om} = -\lambda_+ \ln \lambda_+ - \lambda_- \ln \lambda_-$	$\lambda_\pm$ eigenvalues; Von Neumann entropy formalism for mixed states.	Quantum Communication

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T4	$N_{s,q}/(1-N_{s,q}) \approx \sqrt{3}$	Geometric projection of the Omnium horizon ratio to the SU(3) Color Symmetry coefficient.	Differential Geometry
T5	$\Omega_\Lambda = N_{s,q} + \frac{\Omega_{DM}}{\phi \cdot \pi} \approx 0.6866$	Golden ratio (ϕ) and the logarithmic spiral filter of Channel C gravitational coupling.	Observational Cosmology (Planck)
T7	$\frac{N_{s,q}}{N_{s,c}} = 2\pi G\alpha^2 c\hbar k_B$	Boundary equalization of Hilbert (Quantum) and Minkowski (Classical) spaces (Omnium Bridge).	Grand Unified Theory (GUT)
T8	$\Lambda = \Lambda_P \cdot (N_{s,q}^2)^{2/(1+\alpha)}$	Planck normalized cosmological constant derived via the dimensional regulation of the T7 equation.	Cosmology (Λ Crisis Resolution)
T9	$g_{tt} \cdot g_{rr} = -1$, $G_{\theta\theta} = 0$	de Sitter vacuum solution at the Omnium horizon limit $r_s/r = N_{s,q}$ in the UST metric.	General Relativity
T10	$T_{Om} = \exp[-2\pi N_b C_{cb}] \approx 0.2325$	WKB (Wentzel-Kramers-Brillouin) quantum tunnelling amplitude from Channel Q to Channel C.	Quantum Tunnelling / LIGO
T11	$\frac{p_Q}{p_C} = \frac{N_m}{C_{cm}}\left(1 + \frac{N_b \cdot N_m}{2\pi}\right)$	Differential quadratic intersection of the Blueprint (N_b) and Manifest (N_m) ontological phases.	High Energy Physics (CERN ATLAS/LHCb)
T13	$p_Q/p_{C_P} = T_{11}\left[V_b - \frac{T_{Om} \cdot C_{cb}}{2\pi(3-2N_b)}\right]$	Lindblad Stokes dampening determined by the metric derivative $f'(N_b) = 3 - 2N_b \approx \sqrt{3}$.	CMB Polarization (SPT-CMB)
T16	$\Omega_{DM} + T_{Om} \approx 1/2$	Gravitational shadow (Dark Matter) equalization of Channel C in the active universe.	Astrophysics (Dark Matter)
T17	$r_{Om}/r_S = 1/N_b \approx 1.5784$	Information sealing boundary (Omnium Horizon) outside the Schwarzschild horizon.	Black Hole Physics
T23	$H_{local}/H_{Planck} = 1 + N_b C_{cb}^2$	Reflection of the tunnelling information (Channel C) on the asymmetric expansion acceleration.	Cosmology (Hubble Tension)
T25	$BigBang =$ Channel Inversion	Phase Change process where the T_{Om} amplitude is conserved across universe cycles.	Cyclic Cosmology
T26	$d/L = N_b/C_{cb} \approx 1.727$	Focal depth / Rupture length fracture limit (derived via 52,223 earthquake data).	Seismology / Geophysics
T28	DNA Heterozygous $\approx N_{s,q}$	Active (Channel Q) and archive (Channel C) base coding density in molecular biology.	Molecular Biology / Genomics
T29	$r_{wh}/r_{bh} = N_b/C_{cb} \approx \sqrt{3}$	White/black hole spectrum coefficient at the Omnium horizon limits.	Compact Object Astrophysics
T30	$\eta = 6\pi \cdot N_b \cdot C_{cb} \approx 4.3762$	Channel C projection of the Sommerfeld parameter governing proton-proton tunnelling.	Stellar Nucleosynthesis (Fusion)
T32	$t_H \times N_b = \nu_{Cs-133}$	Equalization of the Hubble Macro-Time to the Cesium-133 micro-oscillation frequency.	Universal Time Standard

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T37	$\Omega_b = \frac{T_{Om}}{\phi \cdot \pi} \cdot \left(1 + \frac{1}{17}\right) \approx 0.0484$	1-loop a_2 (Seeley-DeWitt) CP violation topological shift added to the purely geometric mass budget.	Baryogenesis / QFT
T38	$\Omega_{DM} = T_{Om} + (N_{s,q}/17) \approx 0.2698$	Analytical QFT a_2 coefficient equalization applied to the Dark Matter gravitational coupling.	Astrophysics / Cosmology
T39	$\frac{\dot{\alpha}}{\alpha} = -\ln(1 + \Delta f) \cdot H_0$	Dirac parametric drift derived from the universal cycle residue ($\Delta f = 0.3005$).	Spectroscopy (Quasar Data)
T41	$\frac{d\vec{S}}{dt} = -\left(\frac{T_{Om} C_{cb}}{2\pi(3-2N_b)}\right) L_k \vec{S} L_k^\dagger$	Lindblad Master evolution of Stokes polarization vectors derived from the $SU(3)$ symmetry root.	Open Quantum Systems
T43	$\lim_{n\to\infty} F_{collapse} \geq C_{cb} \cdot T_{Om} \approx 0.0852$	Non-Markovian memory kernel utilizing the Channel C reservoir in $n > 4$ multi-qubit hardware.	Scalable Quantum Processors

3. Academic References and Empirical Databases

- **CERN ATLAS Open Data:** 13 TeV Proton-Proton Collision Data (Run 2, Period D/L), Missing Transverse Energy (MET) and Hadronic Locking ($T11$, $T42$) verifications.
- **ESA Euclid & Planck 2018:** Shape-Entropy Measurements, H_0 Hubble data, $\Omega_\Lambda = 0.6857$ and Dark Matter fraction ($T5, T8, T38$) limit analyses.
- **LIGO / GWOSC:** Gravitational wave deviations (O4a HDF5 data) and T_{Om} tunnelling rate asymptote verifications.
- **DeWitt, B. S. (2003):** *The Global Approach to Quantum Field Theory.* (Theoretical QFT source for the 4D Dirac Operator and Seeley-DeWitt $a_2 = 1/17$ proof).
- **Lindblad, G. (1976):** *On the generators of quantum dynamical semigroups.* (Open system decoherence formalism and $T41$ operator dampening source).
- **Keck/HIRES:** High-redshift ($high - z$) distant quasar absorption spectra ($T39$ fine structure constant drift verification).